NAME: SECTION:
HABIB ID: INSTRUCTOR:

MATH 205 LINEAR ALGEBRA
MIDTERM PART A
SPRING 2026

TIME: 40 MINUTES
TOTAL MARKS: 40

Definitions and Formula Sheet

- A system of n equations with n variables:

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = 0 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = 0 \\ \vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n = 0 \end{cases}$$

- Augmented matrix for the system $AX = B$:

$$\left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} & b_n \end{array}\right]$$

- If A is $m \times r$ and B is $r \times n$, then:

$$AB = \text{Matrix of size } m \times n.$$

- **Transpose of a Matrix:** For a 3×2 matrix:

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}, \quad A^T = \begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \end{bmatrix}.$$

- A matrix is symmetric if $A^T = A$.
- A matrix is skew-symmetric if $A^T = -A$.
- **Elementary Row Operations:**
  1. Multiply a row by a nonzero constant: $(R_i \to cR_i,\ c \neq 0)$
  2. Interchange two rows: $(R_i \leftrightarrow R_j)$
  3. Add a multiple of one row to another: $(R_i \to R_i + cR_j)$
- **Finding A^{-1} using Row Operations:**

$$[A \mid I] \xrightarrow{E.R.O.S} [I \mid A^{-1}].$$

- If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, then $\det(A) = ad - bc$.
- For two square matrices A and B: $\det(AB) = \det(A)\det(B)$.
- For the identity matrix I_n: $\det(I_n) = 1$.
- For an invertible matrix A: $\det(A^{-1}) = \frac{1}{\det(A)}$.
- $\det(A) = \det(A^T)$.
- The determinant of a matrix A can be evaluated using cofactor expansion along any row or column:

$$\det(A) = \sum_j a_{ij}C_{ij},$$

where C_{ij} is the cofactor of a_{ij}.

- The adjoint matrix $\text{adj}(A)$ is the transpose of the cofactor matrix and inverse is given by:

$$A^{-1} = \frac{\text{adj}(A)}{\det(A)}.$$

- **Cramer's Rule:** For a system $AX = B$, the i-th variable is:

$$x_i = \frac{\det(A_i)}{\det(A)},$$

where A_i is obtained by replacing the i-th column of A with B.

- For an upper or lower triangular matrix, the determinant is the product of diagonal elements:

$$\det(T) = t_{11} \cdot t_{22} \cdots t_{nn}.$$

- An n-tuple is an element of $\mathbb{R}^n$. Vector addition and scalar multiplication in $\mathbb{R}^n$ follow:

$$\mathbf{u}+\mathbf{v} = (u_1+v_1, \ldots, u_n+v_n), \quad k\mathbf{u} = (ku_1, \ldots, ku_n).$$

- **Euclidean Inner Product:** For $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$:

$$\mathbf{u} \cdot \mathbf{v} = \sum_{i=1}^{n} u_i v_i.$$

- **Norm and Distance:** The norm (length) and distance are defined as:

$$\|\mathbf{u}\| = \sqrt{\mathbf{u} \cdot \mathbf{u}}, \quad d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|.$$

- **Cauchy–Schwarz Inequality:** For all $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$:

$$|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\|\|\mathbf{v}\|.$$

- **Triangle Inequality:** For all $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$:

$$\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|.$$

- Two vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ are orthogonal if: $\mathbf{u} \cdot \mathbf{v} = 0$. And for orthogonal vectors $\mathbf{u}$ and $\mathbf{v}$:

$$\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2.$$

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Question 1:

(a) If A and B are square matrices of the same size, then prove or disprove the following:

$$\det(A) + \det(B) = \det(A + B)$$ **[4 Marks]**

(b) **Prove**: If $\det(A) = 1$ and all the entries in A are integers, then all the entries in A^{-1} are integers.

[6 Marks]

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Question 2: **Prove**: The square matrix A is invertible if and only if $\det(A) \neq 0$. **[10 Marks]**

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Question 3: For which values of a does the following system have: (a) no solutions, (b) exactly one solution, or (c) infinitely many solutions? **[10 Marks]**

$$\begin{aligned} x + 2y &= 4 \\ 3x + ay &= 6 \end{aligned}$$

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Question 4: Let A be a n x n invertible matrix. Then prove that: **[10 Marks]**

$$(A^T)^{-1} = (A^{-1})^T.$$

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ROUGH WORK

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Question 1: (a) Briefly explain what you understand by a *consistent system of equation*s. **[2 Marks]**

(b) Find the conditions that the b's must satisfy for the following system to be consistent:

$$\begin{aligned} 4x_1 - 6x_2 &= b_1 \\ 2x_1 - 3x_2 &= b_2 \end{aligned}$$

[8 Marks]

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Question 2: **Prove:** If A and B are square matrices of the same size, then:

$$\det(A\,B) = \det(A)\,\det(B)\,.$$ **[10 marks]**

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Question 3: If ***u, v,*** and ***w*** are vectors in R^n, then prove the following:

(a) $(\boldsymbol{u}+\boldsymbol{v}).\boldsymbol{w}=\boldsymbol{u}.\boldsymbol{w}+\boldsymbol{v}.\boldsymbol{w}$ [**4 marks**]

(b) $\|\boldsymbol{u}+\boldsymbol{v}\| \leq \|\boldsymbol{u}\|+\|\boldsymbol{v}\|$ **[6 marks]**

Question 4: **(a)** State the conditions for a matrix to be in *reduced row echelon* form. **[4 marks]**

(b) Now use the theorem: "A matrix A is invertible if and only if its reduced row echelon form is the identity matrix" to find the inverse of the following matrix: **[7 marks]**

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

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